

Conduct repeated chance experiments and simulations, using digital tools to determine probabilities for compound events, and describe results

Learning intention

- conduct repeated chance experiments and simulations, using digital tools to determine probabilities for compound events, and describe results.

Teach and model

- using online simulation software to conduct probability simulations to determine in the long run if events are complementary.
- using digital tools, including generative artificial intelligence, to conduct probability simulations involving compound events.
- using a random number generator and digital tools, including generative artificial intelligence, to simulate rolling 2 dice and calculating the difference between them.

Check understanding

- Explain the main idea and give one accurate example.
- Show the idea in a second way: with words, a model, a diagram or symbols.
- Apply the idea to a short unfamiliar situation and justify the result.

Teaching cautions

- Ask the student to explain their choice, not only state an answer.
- Check that key vocabulary is understood in a new example.
- Mix representations so the skill is transferred rather than copied.

Quick lesson sequence

- Connect to prior knowledge and introduce the key vocabulary.
- Model one clear example, then solve a second example together.
- Finish with an independent check and a short student explanation.